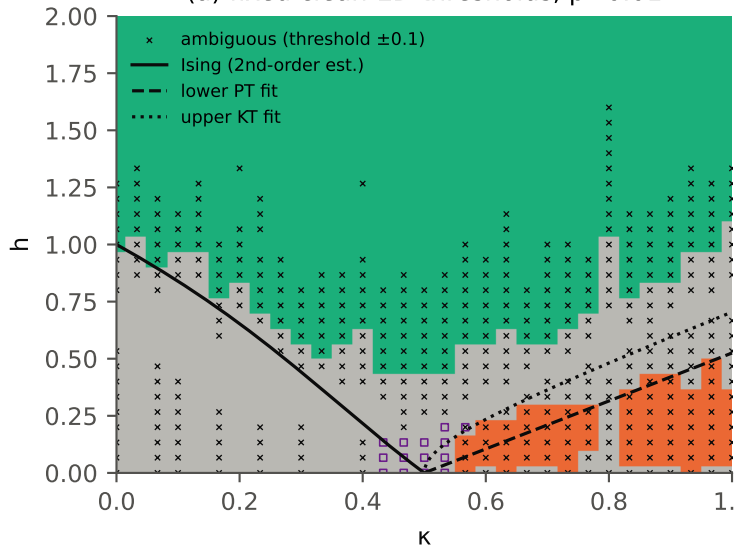
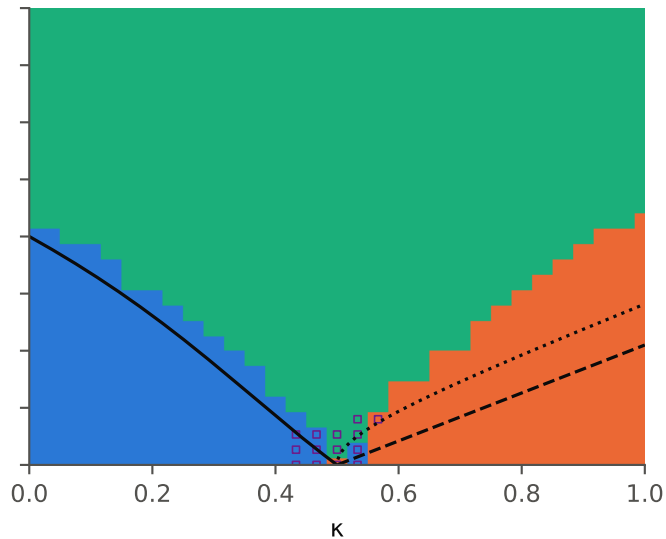


HVA circuit N=8 PBC, L=4 (128 CNOTs, 128 target channels), $p=0.01$

(a) fixed clean-ED thresholds, $p=0.01$



(b) relative dominance $\text{argmax}(\text{O}_F, \text{O}_A, m_{x^2})$, $p=0.01$



ferro antiphase paramagnet unpolarised (order & m_x low)

Fixed clean-optimised parameters for all p . Purple squares: clean preparation fidelity < 0.9 . x : label changes for thresholds ± 0.1 . Black dots in (b): largest signal < 0.05 .